Laxmi Karthikeya Gollapudi

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Summary

Second-semester Engineering student focused on AI applications and interdisciplinary scientific exploration. Experienced in developing **Computer Vision** and **IoT projects**, with strong interests in **research, technical writing**, and gaining exposure across diverse Computer Science fields.

Education

B.Tech in Computer Science & Engineering
PES University, 2025-2029

Skills

- **Languages:** Python, C, HTML, CSS, JavaScript
- **AI:** Machine Learning, Regularisation models, Anomaly detection models, Computer Vision
- **IoT & Integration:** Sensors, Data Acquisition, Real-time Monitoring
- **Libraries & Tools:** OpenCV, YOLOv8, Kivy, Firebase, pandas, NumPy, matplotlib, scikit-learn, glob, JSON, PyTorch
- **Other:** Data Analysis, Git/GitHub, Technical Writing

Experience and Projects

Linear Algorithm for Data Reconstruction for the ATLAS Tile Calorimeter (HL-LHC) link

- Completed a technical warm-up problem related to energy signal reconstruction for the ATLAS detector
- Implemented a **linear regression model** to reconstruct calorimeter energy signals from detector data
- Evaluated reconstruction accuracy and performance of the algorithm

Tech Stack: Python, NumPy, scikit-learn, matplotlib

Anomaly Detection Using prmon Synthesized Data link

- Generated synthetic process resource monitoring time-series data using **prmon**
- Developed an **autoencoder**-based anomaly detection model to identify abnormal resource usage patterns
- Used reconstruction error thresholds to detect anomalies in mixed normal and anomalous datasets

Tech Stack: Python, PyTorch, NumPy, pandas, matplotlib, prmon

Comparative Study on Autoencoders and Their Variants link

- Implemented and compared multiple **autoencoder** architectures to study representation learning and reconstruction behaviour
- Benchmarked models using the **MNIST** dataset
- Evaluated performance across reconstruction accuracy and latent representation quality

Tech Stack: Python, PyTorch, NumPy, matplotlib

Skill-Based Multiplayer Gaming System (Rock–Paper–Scissors) [link]

- Trained a **YOLOv8** computer vision model for real-time gesture recognition
- Developed an automated scoring system using **OpenCV** for precise hand detection and validation
- Designed a skill-based multiplayer experience to enhance gameplay fairness and responsiveness

Tech Stack: Python, YOLOv8, OpenCV, NumPy

Crowdsourced Pothole Detection Mobile App b [link]

- Developed a mobile application using **Kivy** and **Firebase**
- Used accelerometer and gyroscope data to detect road anomalies in real time
- Implemented a threshold-based filtering algorithm to distinguish potholes from normal driving behaviour

Tech Stack: Python, Kivy, Firebase, Mobile Sensors (Accelerometer, Gyroscope)

Portfolio Website [link]

- Built a personal portfolio website using a Static Site Generator
- Customized the website using **HTML, CSS,** and **JavaScript**
- Used the site to showcase technical projects and blog posts

Tech Stack: HTML, CSS, JavaScript, Static Site Generator (SSG)

ACM Applied Internship Experience Program (AIEP)

- Selected for the **ACM Applied Internship Experience Program**
- Participating in a collaborative **research study** comparing regularization techniques in machine learning
- Evaluating models based on reconstruction quality, latent representation, and generalization

Tech Stack: Python, PyTorch, NumPy

Certifications

- **HackerRank Certification – Python Basics**

Links

- **Website:** https://karthikgollapudi.vercel.app/
- **GitHub:** https://github.com/L-Karthik-G
- **Medium Blog:** https://medium.com/@glkarthik27
- **HackerRank Certificate:** https://www.hackerrank.com/certificates/5a6c02cda3c4
